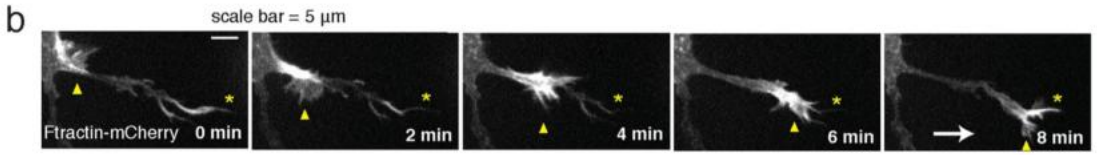
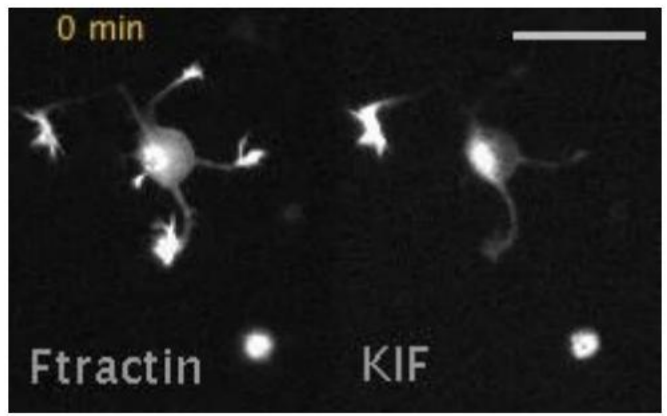
Actin waves accompany bursts of neurite growth

- **Actin waves** are traveling regions of **filamentous actin** that move outward through immature **neurites**.
  - A **neurite** is a growing projection that may become an axon or dendrite.
  - Researchers tracked fluorescent F-actin waves in cultured **rat hippocampal neurons**.
- A passing wave briefly **widens the neurite shaft** and rearranges its **cytoskeleton**.
  - **Microtubules** can extend into the wider region and provide tracks for cargo.
  - **Kinesin** moves cellular cargo along these microtubules toward the neurite tip.
- **Neurite extension occurs in bursts** when actin waves reach growing tips.
  - A neurite may remain stable and then lengthen rapidly as a wave arrives.
  - Waves reach different branches at different times, producing uneven growth.



Winans et al. (2016), Fig. 1b. Developing neurites, actin, and intracellular transport.



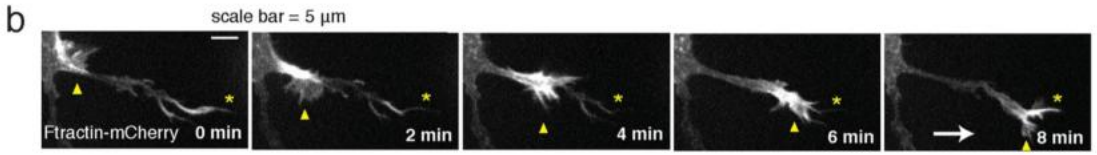
Winans et al. (2016), Video 5. Academic: Neurite growth and transport.

Actin waves accompany bursts of neurite growth

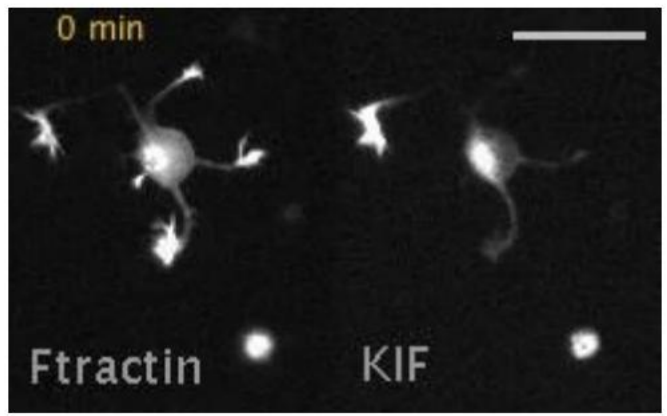
Developing neurons extend processes called **neurites**, which may later become axons or dendrites. **Actin waves** are moving regions rich in **filamentous actin**. Fluorescent labeling allowed Winans and colleagues to track these waves in cultured rat hippocampal neurons.

A passing actin wave briefly **widens the neurite shaft** and changes the local **cytoskeleton**. **Microtubules** extend into the wider region, providing structural support and tracks for intracellular cargo. **Kinesin motor proteins** move that cargo along microtubules toward the neurite tip.

Neurites do not always grow steadily. **Extension occurs in bursts** associated with arriving actin waves. One process may remain stable before lengthening rapidly, while another grows at a different time. Waves reaching individual branches at different times produce different patterns of neurite growth.



Winans et al. (2016), Fig. 1b. Developing neurites, actin, and intracellular transport.



Winans et al. (2016), Video 5. Academic: Neurite growth and transport.